

BINARY NUMBERS

Count · Learn · Explore

Grade 6 · AI & Digital

1. WHAT ARE NUMBERS?

Numbers Are Everywhere Around Us!

Think about counting your fingers, steps you take, or toys you have. Numbers help us count, measure, and describe things in our world!

We Use Numbers Every Single Day

Numbers help us understand and describe the world around us. From counting friends to measuring distances, numbers are our helpful tools!

2. DIFFERENT WAYS TO COUNT

Decimal System

We use base 10 every day with digits 0–9.
Count your fingers — that's decimal!

Binary System

Computers use base 2 with only 0 and 1.
Simple but powerful!

Side by Side

Decimal: 0, 1, 2, 3, 4, 5 ... Binary: 0, 1, 10, 11, 100, 101 ...

3. WHAT IS BINARY?

Binary: The Language of 0s and 1s

Binary is a number system that uses only two digits: 0 and 1. Every number, letter, and image in a computer is made from combinations of these two simple digits!

Why Computers Love Binary

Computers use binary because they work like light switches — either ON (1) or OFF (0). This simple on/off system makes computers fast and reliable!

4. HOW BINARY WORKS

Each digit is called a BIT	Bits represent powers of 2	Place values: 1, 2, 4, 8, 16, 32 ...
--------------------------------------	--------------------------------------	--

Example: Binary 101

Breaking it down by place value ...

Calculation

$$1 \times 4 + 0 \times 2 + 1 \times 1 = 5$$

5. COUNTING IN BINARY (0 – 15)

Decimal	Binary	Decimal	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111

6. WHY DO COMPUTERS USE BINARY?

Simple and Reliable for Electronic Circuits

Electronic circuits work like tiny switches that can only be ON or OFF. Binary's two digits (0 and 1) match perfectly with these switches, making computers fast and dependable!

Easy to Store and Process Information

With only two options to remember, computers make fewer mistakes! Binary is less error-prone, helping your devices work smoothly every time you use them.

7. CONVERTING BETWEEN DECIMAL AND BINARY

Decimal → Binary

Step 1	Divide the decimal number by 2
Step 2	Write down the remainder (0 or 1)
Step 3	Repeat with the quotient until you reach 0
Step 4	Read the remainders backwards — that's your binary number!
Example	13 → 1101

Binary → Decimal

Step 1	Write down powers of 2 under each bit (right to left: 1, 2, 4, 8 ...)
Step 2	Multiply each bit by its place value
Step 3	Add all the results together
Example	1010 → $1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 10$

8. BINARY IN ACTION: AI & DIGITAL DEVICES

■ Voice Assistants

Binary helps AI assistants understand your voice and respond to questions!

■ Games & Apps

Every game and app uses binary code to create graphics and sounds.

■ Smartphones

Your phone uses billions of binary bits to store photos, messages, and more!

9. FUN BINARY FACTS

■ **Computers use millions of bits every second!**

■ **A byte equals 8 bits of data**

■ **Binary creates images, sounds & text**

■ **Binary is the language of all digital devices**

10. QUIZ TIME! ■

Think carefully about what you learned. Answer the questions below!

Q1. What two digits are used in binary?

Answer:

Q2. What decimal number is binary 1001?

Answer:

Q3. What do we call a single binary digit?

Answer:

Q4. How many bits are in one byte?

Answer:

Q5. Convert decimal 6 to binary.

Answer:

11. RECAP: WHAT WE LEARNED

■ **Binary Uses 0 and 1**

Just two digits create all digital information!

■ **Computers Count Binary**

Machines store and process everything using bits.

■ **Easy Conversions**

Decimal to binary and back again — you can do it!

■ **Binary Is Everywhere**

Phones, games, AI — all digital devices use binary!

FURTHER RESOURCES

Resource	Description	Website
Code.org	Free Coding Games	code.org
Scratch	Kids Programming	scratch.mit.edu
Khan Academy	Binary Lessons	khanacademy.org

Keep Exploring! Try counting in binary at home using your fingers. Curiosity is the best way to learn! ■